

MOSTAFA SAMIR KAMEL

Front-End Developer | MERN Stack Developer | AI & Data Science Graduate

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PROFESSIONAL SUMMARY

Front-End Developer and 2026 graduate in Artificial Intelligence & Data Science with hands-on experience across both the MERN (MongoDB, Express.js, React, Node.js) and MEAN (MongoDB, Express.js, Angular, Node.js) stacks. Skilled in building responsive, production-ready React.js and Next.js applications, RESTful APIs, and full-stack JavaScript solutions, alongside experience developing AI models and embedded systems. Strong analytical and problem-solving skills with a passion for creating user-friendly web applications and contributing to innovative development teams.

EDUCATION

BSc in Artificial Intelligence & Data Science — Beni Suef National University, College of Computing and Artificial Intelligence
Graduated 2026

GPA: 3.35 / 4.00

EXPERIENCE

Machine Learning Engineer Intern — Technocolabs Softwares Inc.

Mar 2025 – May 2025

- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) on real-world datasets
- Applied classification and clustering algorithms, improving model accuracy through data optimization techniques
- Collaborated in an Agile environment, contributing to improved workflow efficiency and model performance

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C, C++, C#, Java, Embedded C, HTML5, CSS3

Front-End: React.js (17+), Next.js, Redux Toolkit, React Router, React Hook Form, Formik, Zod/Yup, Tailwind CSS, Bootstrap 5, Material-UI, Ant Design, Sass

Back-End & APIs: Node.js, Express.js, MongoDB, MySQL, PostgreSQL, SQL Server, RESTful APIs, Flask, FastAPI, JWT, Mongoose

MERN Stack: MongoDB, Express.js, React, Node.js — full-stack JavaScript application development (primary stack, used across production projects)

MEAN Stack: MongoDB, Express.js, Angular, Node.js — full-stack JavaScript application development (NTI 6-week training)

AI & Machine Learning: scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, OpenCV, YOLO, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly

Deep Learning: ANN, CNN, RNN, ResNet, DenseNet, U-Net, Transformers (NLP & CV), AutoEncoders, GANs

Embedded Systems: AVR (ATmega32), ARM, Arduino, GPIO, ADC, Interrupts, EEPROM, Timers, RTOS/FreeRTOS, UART, I2C, SPI, AUTOSAR

Cloud & DevOps: Vercel, Netlify, GitHub Pages, Hostinger, GitHub Actions, Jenkins, CI/CD

Tools: Git & GitHub, GitLab, VS Code, Postman, Chrome DevTools, npm, yarn

Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), SDLC, Agile & Waterfall, Requirements Engineering (SRS), Operating Systems, Computer Networking

PROJECTS

Flagship Graduation Project

RAFIQ — AI-Powered Autism Therapy Platform | [Live Demo](#)

Next.js, React, Redux Toolkit, Node.js, Express, MongoDB + 5 Dockerized FastAPI AI microservices (DeepFace, MediaPipe, PyTorch Arabic ASR, OpenCV, DistilGPT-2). Arabic-first ASD therapy & progress-management platform.

Next.js & React Applications (Live on Vercel)

Rayan (Map & Geo-Location App) | [GitHub](#) | [Live Demo](#)

Next.js 16, React 19, Redux Toolkit, React Leaflet, Zod, i18n — geospatial tracking application.

Shop Mart (E-Commerce) | [GitHub](#) | [Live Demo](#)

Next.js 16, React 19, Tailwind CSS, Shadcn UI — dynamic product catalog, filters, stateful cart.

Recipe Studio | [GitHub](#) | [Live Demo](#)

Next.js 16, TypeScript, Tailwind CSS — recipe discovery with server-side rendering and dynamic routing.

Northbench Academy | [GitHub](#) | [Live Demo](#)

Next.js 14, TypeScript, Dark Mode — marketing & admissions portal with filterable course catalog.

Route Posts (Social Feed) | [GitHub](#) | [Live Demo](#)

React (Vite), TanStack Query, Axios, PrimeReact — social feed with auth, likes, comments, bookmarks.

Social Network App | [GitHub](#) | [Live Demo](#)

React (Vite), Flowbite, React Query, Zod, React Hook Form — follow feeds, profile dashboard, secure forms.

Bakery Catalog App | [GitHub](#) | [Live Demo](#)

React (Vite), Bootstrap, Axios, React Toastify — bakery store with cart management and notifications.

Saraha (Anonymous Feedback Platform – Frontend) | [GitHub](#) | [Live Demo](#)

Next.js 16, React 19, Framer Motion, QR Code React — anonymous messaging client with QR profiles.

FreshCart (Grocery E-Commerce) | [GitHub](#) | [Live Demo](#)

Next.js 15, React 19, Tailwind CSS — grocery storefront with dynamic catalog and cart management.

React Movie Explorer | [GitHub](#) | [Live Demo](#)

React 19, Vite, React Router v7, Context, Axios, TMDb API — movie SPA with favorites and pagination.

React Portfolio Dashboard | [GitHub](#) | [Live Demo](#)

React 19, Vite, React Router v7, Bootstrap — interactive single-page portfolio with client-side routing.

Node.js Backend

Saraha Backend (Secure Messaging API) | [GitHub](#)

Node.js, Express, MongoDB/Mongoose, Redis, Argon2, Joi, Nodemailer/Resend, Cloudinary, Jest & Supertest — production-grade REST API with caching, security headers, and test suites.

Static Web Applications (Live on GitHub Pages)

NutriPlan (Nutrition & Fitness Tracker) | [GitHub](#) | [Live Demo](#)

HTML5, JS (ES6 Modules), Tailwind CSS, Plotly.js — food scanner, recipe search, nutrition charting.

Muddaber (Personal Finance Tracker) | [GitHub](#) | [Live Demo](#)

HTML5, Vanilla JS, Bootstrap RTL — Arabic budget tracker with full RTL support.

DJI Mavic 4 Pro Landing Page | [GitHub](#) | [Live Demo](#)

HTML5, CSS3, Vanilla JS — marketing page with scroll transitions and specs accordion.

GameArena (Gaming Portal) | [GitHub](#) | [Live Demo](#)

HTML5, CSS3, JavaScript — dark-themed portal with dashboard grid and countdowns.

CodeAlpha Image Gallery | [GitHub](#) | [Live Demo](#)

HTML5, CSS3, Vanilla JS, Bootstrap — portfolio gallery with search filters and zoom lightbox.

The UX Review Blog | [GitHub](#) | [Live Demo](#)

HTML5, CSS3, responsive typography — readability-focused blog with article grids and tag sorting.

Pure JavaScript & Web API Applications

Product Management CRUD System | [GitHub](#) | [Live Demo](#)

HTML5, Vanilla JS, LocalStorage, Bootstrap — inventory tool with tax/discount calculation and search.

Weather Forecasting App | [GitHub](#) | [Live Demo](#)

HTML5, Vanilla JS, Weather REST API — real-time forecaster with geolocation support.

Asynchronous Movie Browser | [GitHub](#) | [Live Demo](#)

HTML5, Vanilla JS, TMDb API — lightweight movie search with async fetches and live filters.

Local Authentication System | [GitHub](#) | [Live Demo](#)

HTML5, Vanilla JS, LocalStorage, Bootstrap — register/login system with regex validation.

Random Quote Generator | [GitHub](#) | [Live Demo](#)

HTML5, Vanilla JS, Quotes API — quote engine with dynamic backgrounds and social sharing.

DevFolio Template | [GitHub](#) | [Live Demo](#)

HTML5, CSS3, JavaScript, Bootstrap — developer portfolio with typing animations and skill indicators.

Mealify Landing Page | [GitHub](#) | [Live Demo](#)

HTML5, CSS3, Grid/Flexbox — restaurant portal with gallery, menus, and testimonials.

Premium 3D & Showcase Sites

Apple iPhone 15 Pro Showcase | [Live Demo](#)

High-fidelity 3D replica marketing page with scroll-driven animations and device orbit controls.

Interactive 3D WebGL Portfolio | [Live Demo](#)

Immersive portfolio with WebGL/Three.js visuals and parallax scene transitions.

Closca Detox Juice Landing Page | [Live Demo](#)

Premium product page with vibrant visual contrasts and CSS animations.

AI & Machine Learning Projects

Customer Segmentation (Unsupervised Learning) | [GitHub](#)

K-Means clustering, Elbow Method, Seaborn — segments retail shoppers by income and spending habits.

Fraud Detection Classification | [GitHub](#)

Feature engineering, imbalanced-data handling, F1/Precision/Recall evaluation of fraud detection models.

Hand Gesture Recognition & Tracking | [GitHub](#)

MediaPipe landmark extraction, SVM classifier, OpenCV real-time webcam inference.

Handwritten Digit Recognition | [GitHub](#)

CNN trained on MNIST, deployed via Flask API with OpenCV pre-processing.

HR Analytics (Employee Attrition Predictor) | [GitHub](#)

Scikit-learn Random Forest Classifier predicting attrition from the IBM HR Analytics dataset.

Startup Funding Analysis | [GitHub](#)

EDA mapping global startup funding cycles, VC trends, and industry distributions with Pandas/Seaborn.

Spam Email Filter (NLP) | [GitHub](#)

Tokenization, TF-IDF vectorization, and Naive Bayes classification for spam detection.

Embedded Systems Projects (ATmega32)

Real-Time Clock | [GitHub](#) | [LinkedIn Demo](#)

Timer2 asynchronous overflow mode with 32.768kHz crystal, Keypad adjustment, 7-segment + LCD display.

Traffic Light Controller | [GitHub](#) | [LinkedIn Demo](#)

Timer0 CTC mode intersection controller with hardware interrupts for emergency vehicle priority.

Electronic Voting Machine | [GitHub](#) | [LinkedIn Demo](#)

Button polling with hardware debounce and LCD interface for real-time vote tallies.

Digital Calculator | [GitHub](#) | [LinkedIn Demo](#)

4x4 matrix Keypad driver with LCD parsing algebraic operations.

Temperature Control System | [GitHub](#) | [LinkedIn Demo](#)

LM35 sensor via ADC, LCD status display, relay-driven cooling fan activation.

Counter – 7-Segment Multiplexing | [GitHub](#) | [LinkedIn Demo](#)

Time-division multiplexing of multiple 7-segment displays via Timer0 CTC interrupts.

Safe Lock System | [GitHub](#) | [LinkedIn Demo](#)

EEPROM-persisted passcode validation with Keypad input and alarm states.

Relay & Power Control Drive | [GitHub](#) | [LinkedIn Demo](#)

GPIO-driven relay switching for isolated high-current AC/DC load control (e.g., 12V lamp).

LCD Animations Engine | [GitHub](#) | [LinkedIn Demo](#)

Custom CGRAM graphics for icons, animated progress bars, and scrolling transitions.

Seven Segment Display Driver | [GitHub](#) | [LinkedIn Demo](#)

Modular HAL driver for common anode/cathode 7-segment displays in C.

LED, Button & Buzzer Peripheral Driver | [GitHub](#) | [LinkedIn Demo](#)

I/O driver integrating pull-up buttons, LED grids, and debounced passive buzzers.

TRAINING & CERTIFICATIONS

Web Development: Full-Stack MEAN Development (6 Weeks) – National Telecommunication Institute (NTI); Frontend Development (React.js) – Route Academy; Backend Development (Node.js) – Route Academy

AI & Data Science: Computer Vision Engineer – NTI; Generative AI – Sprints; Machine Learning – Sprints; Intro to Deep Learning – Kaggle; Intro to Machine Learning – Kaggle; AI & Applications – Zewail City; AI & Data Science – DotPy Academy; Data Analysis – Almdrasa

Programming & CS: Python Programming – Microsoft; Programming Fundamentals – Beni-Suef University; OOP & Data Structures – Tasabuq; Computer Science Foundations – University of Delaware

Embedded Systems: Embedded Systems (AVR) – ITI; Embedded Systems (AVR) – Eng. Mohamed Zaghloul; Embedded Systems (ARM) – Eng. Ahmed Bahaa

Events: DevFest – GDSC EELU Beni-Suef

SOFT SKILLS

Problem Solving | Team Collaboration | Attention to Detail | Continuous Learning | Adaptability | Time Management | Communication

LANGUAGES

Arabic: Native English: Intermediate (Reading & Writing)